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Deformation Vector Field Curl as a Method for Detecting Lung Image Registration Errors

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INTRODUCTION

Deformable image registration is a key component of many free-breathing lung function studies, including breathing motion modeling for radiation therapy. DIR often fails locally while providing an overall good registration and quantifying and localizing those failure regions is time consuming and qualitative. Our proposed approach may provide an efficient and quantitative detection of poorly registered regions within the lungs. Future work could lead to even subtle DIR defects being detectable, which could lead to further (and unknown) improvements in breathing motion modeling.

AIM

Image registration poses a difficult problem in many multi-scan lung imaging modalities such as 5DCT. Patients that breath too quickly or too irregularly can often cause image registration results to be unusable, and detecting issues with registration results is often time consuming. We generally assume that deformation vector fields (DVFs) of free-breathing lung CT scans will have limited volumetric rotation, while erroneously registered regions of interest may show greater than physical rotation. We propose using the curl of the DVFs to detect regions in the lungs with poor registration quality.

METHOD

Five patients that had been scanned using a 5DCT scanning protocol (25 back-to-back helical CT scans) were selected to evaluate this approach. The 25 scans were registered using the dense displacement sampling (deeds) or pTV methods. The DVF curls were analyzed with the nominal voxel resolution (1 x 1 x 1 mm^3) and by averaging the vector fields over 2 and 4 voxel cubic regions. Each scale was evaluated with the goal of determining a threshold curl magnitude that would detect poor image registration. We compared regions of large curl magnitudes to visual inspections of the deformable image registration (DIR) quality.

RESULTS

Examination of the curl fields showed that the unaveraged and 2-voxel average DVFs were unable to differentiate between good and poor registrations. However, a threshold of 0.8 for the 4-voxel cubic ROI averaged DVFs was useful in detecting regions with poor image registration.

- Figure 1 shows the 4x4x4 mm^3 volume averaged curl for one of the registered image pairs alongside the deformed and reference images. Blurring and unphysical deformations are clear in this example.
- Figure 2 shows the curl for another image alongside the reference image and the registered image once more. The registration process seems to have produced few errors with a correspondingly low curl.

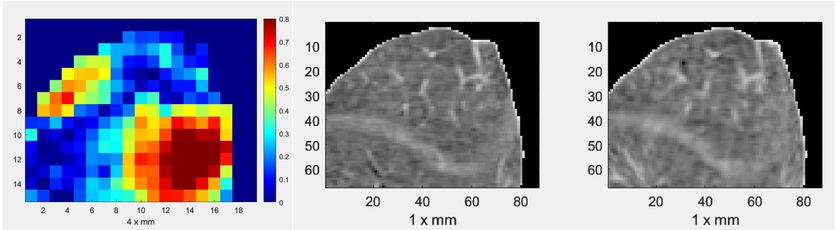


Figure 1: *The curl field (left) is compared against the reference image (center) and the registered image (right). Areas of high curl are more red than areas of low curl, which are more blue. The target image, taken from the scanner, seems to have artefacts present which may have affected the registration process – the registered image contains "ghost" structures and is missing structures which should be there.*

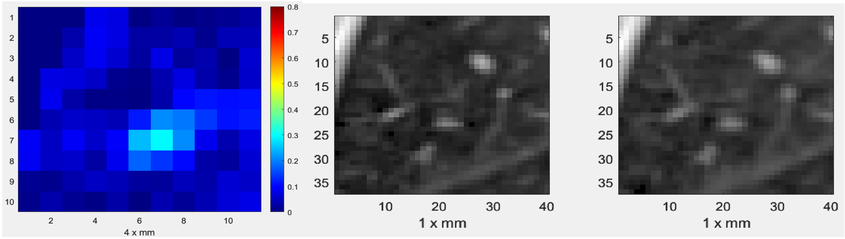


Figure 2: *The curl field (left) is compared against the reference image (center) and the registered image (right). Areas of high curl are more red than areas of low curl, which are blue. The curl scale here is the same as the scale above, indicating far lower overall curl for this registration. Accordingly, the registered image closely matches the reference image.*

CONCLUSIONS

This method of poor registration detection shows promise in providing an automated method for detecting poor quality lung CT image registration results, which would be useful for processes such as 5DCT that require the registration of many free-breathing CT scans. Further research is necessary to optimize both the DVF curl analysis scale and the detection threshold.

REFERENCES

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Can be Vancouver style i.e. 1 **Meyer J-P et al**. The treatment of high-grade superficial bladder cancer and carcinoma in situ with BCG – a questionnaire survey of Consultant practice in England and Wales. *Urol Oncol 2002; 2;: 77-80*

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